

MIMO Techniques in 5/6G (F-js851-3*)

Technical Abstract

CSI-Feedback-Driven Link Adaptation in 5G NR Downlink MIMO: A Link-Level Simulation Study

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Background and Motivation

Channel-state-information (CSI) feedback is essential for downlink multiple-input multiple-output (MIMO) transmission in 5G New Radio (NR). Based on channel measurements at the user equipment (UE), the reported rank indicator (RI), precoding matrix indicator (PMI), and channel quality indicator (CQI) allow the base station to adapt the number of spatial layers, precoding matrix, modulation scheme, and target code rate. However, CSI reports are only useful while they remain representative of the channel at the time of transmission. Increasing the reporting frequency can improve CSI freshness, but also introduces additional uplink feedback overhead and UE processing cost. This project investigates the link-level impact of MIMO configuration, propagation conditions, mobility, and CSI reporting periodicity on downlink throughput.

Methodology

A 5G NR downlink link-level simulation was developed using MATLAB 5G Toolbox. The baseline configuration uses a 3.5 GHz carrier, 40 MHz bandwidth, 30 kHz subcarrier spacing, a 4×4 MIMO antenna configuration, and a CDL-C channel with 300 ns delay spread. CSI feedback is generated using periodic CSI-RS measurements with practical channel estimation. Wideband CQI and PMI reporting, adaptive RI selection, and a Type-I single-panel codebook are used. The main experiments measure successfully delivered PDSCH throughput without HARQ retransmission, allowing the first-transmission impact of CSI-driven link adaptation to be isolated.

The simulation framework is first checked against externally reported fixed-MCS BLER curves. Subsequent experiments examine the baseline link-adaptation behaviour, fixed-rank controls, antenna-configuration and CDL-profile sweeps, and mobility scenarios corresponding to maximum Doppler shifts of 10 Hz, 100 Hz, and 390 Hz. Finally, CSI reporting periods of 4, 10, 64, and 160 slots are compared under each mobility condition.

Key Results and Interpretation

Under the baseline CDL-C 4×4 configuration, throughput increases with SNR and saturates at approximately 164 Mbps. CSI statistics show that the link-adaptation mechanism changes the selected CQI, modulation scheme, code rate, and transmission rank as the channel quality improves. The selected RI increases from one to two layers, but does not exceed two under the baseline channel conditions. Fixed-rank controls confirm that restricting transmission to a single layer limits the high-SNR throughput, while allowing more than two layers provides little additional benefit in this scenario.

Increasing mobility right shifts the throughput curve towards higher SNR. For example, the SNR required to reach 80 Mbps increases from 7.7 dB at 10 Hz to 10.5 dB at 390 Hz. As mobility increases, the channel changes more rapidly, making the reported CSI outdated by the time it is used for transmission. This reduces the accuracy of link-adaptation decisions and leads to lower throughput.

CSI reporting periodicity mainly affects the transition region of the throughput curve rather than the high-SNR saturation level. Relative to the 10-slot baseline, long reporting periods introduce measurable SNR penalties. At 10 Hz, increasing the period to 160 slots produces a 2.4 dB penalty at the selected reference throughput. Reducing the period to 4 slots provides only a modest improvement, with the largest observed gain being approximately 0.5 dB at 100 Hz. Therefore, more frequent CSI reporting is only beneficial under certain mobility and SNR conditions.

Conclusion

The results demonstrate that downlink throughput is jointly determined by channel quality, spatial structure, mobility, and the freshness of CSI feedback. More frequent reporting can reduce the SNR required to achieve a given throughput, but the benefit is scenario-dependent and becomes limited when other effects, such as processing delay or rapid channel variation, dominate. The study provides a link-level basis for future work on adaptive CSI-reporting strategies that explicitly balance downlink gain against uplink overhead and UE power consumption.